ITA14 - ETHICAL HACKING
EX - 8

Execute and display the results for the following Nmap service version and OS detection commands in Kali Linux operating systems:
i) detect the version of services running
ii) Aggressive scan
iii) Detect operating system of the target

Aim
To execute and display the results for the following Nmap service version and OS detection commands in Kali Linux operating systems: i) detect the version of services running ii) aggressive scan iii) detect operating system of the target

Procedure
- Open (Cloud Shell Editor)
- i) detect the version of services running

1. sudo apt update -- press Enter
2. sudo apt install nmap -y -- press Enter
3. nmap --version -- press Enter
4. nmap -sV scanme.nmap.org -- press Enter

```
┌──(kali㉿kali)-[~]
└─$ sudo apt update
[sudo] password for kali:
Hit:1 http://http.kali.org/kali kali-rolling InRelease
1554 packages can be upgraded. Run 'apt list --upgradable' to see them.

┌──(kali㉿kali)-[~]
└─$ sudo apt install nmap -y
nmap is already the newest version (7.991+dfsg-1kali1).
nmap set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1554

┌──(kali㉿kali)-[~]
└─$ nmap --version
Nmap version 7.991 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.9 openssl-3.6.2 libssh2-1.11.1 libz-1.3.2 libpcre2-10.48 libpcap-1.10.6 nmap-libdnet-1.18
.0 ipv6
Compiled without:
Available nsock engines: epoll poll select

┌──(kali㉿kali)-[~]
└─$ nmap -sV scanme.nmap.org
Starting Nmap 7.991 ( https://nmap.org ) at 2026-10-27 09:12 -0400
Warning: Hostname scanme.nmap.org resolves to 2 IPs. Using 45.33.32.156.
Warning: 45.33.32.156 giving up on port because retransmission cap hit (10).

zsh: suspended  nmap -sV scanme.nmap.org

┌──(kali㉿kali)-[~]
└─$ nmap -sV scanme.nmap.org
Starting Nmap 7.991 ( https://nmap.org ) at 2026-10-27 09:21 -0400
Warning: Hostname scanme.nmap.org resolves to 2 IPs. Using 45.33.32.156.
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.018s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 965 filtered tcp ports (no-response), 32 closed tcp ports (reset)
PORT     STATE SERVICE    VERSION
22/tcp   open  ssh        OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp   open  http       Apache httpd 2.4.7 ((Ubuntu))
9929/tcp open  tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 37.38 seconds
```

- ii) aggressive scan

1. nmap -A scanme.nmap.org -- press Enter

```
┌──(kali㉿kali)-[~]
└─$ nmap -A scanme.nmap.org
Starting Nmap 7.991 ( https://nmap.org ) at 2026-10-27 09:47 -0400
Warning: Hostname scanme.nmap.org resolves to 2 IPs. Using 45.33.32.156.
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.030s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 970 filtered tcp ports (no-response), 27 closed tcp ports (reset)
PORT     STATE SERVICE    VERSION
22/tcp   open  ssh        OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|   256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_  256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
80/tcp   open  http       Apache httpd 2.4.7 ((Ubuntu))
|_http-favicon: Nmap Project
|_http-title: Go ahead and ScanMe!
|_http-server-header: Apache/2.4.7 (Ubuntu)
9929/tcp open  tcpwrapped
Device type: bridge|VoIP adapter|general purpose
Running (JUST GUESSING): Oracle Virtualbox (90%), Slirp (90%), AT&T embedded (88%), QEMU (86%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (90%), AT&T BGW210 voice gateway (88%), QEMU user mode ne
twork gateway (86%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 80/tcp)
HOP RTT     ADDRESS
1   0.33 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 80.97 seconds
```

- iii) detect operating system of the target

1. sudo nmap -O scanme.nmap.org -- press Enter

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -O scanme.nmap.org
[sudo] password for kali:
Starting Nmap 7.991 ( https://nmap.org ) at 2026-10-27 10:05 -0400
Warning: Hostname scanme.nmap.org resolves to 2 IPs. Using 45.33.32.156.
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.019s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 961 filtered tcp ports (no-response), 36 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
80/tcp   open  http
9929/tcp open  nping-echo
OS fingerprint not ideal because: Didn't receive UDP response. Please try again with -sSU
No OS matches for host

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 34.99 seconds
```

Result:
Thus, the Nmap commands for service version detection (-sV), aggressive scanning (-A), and operating system detection (-O) were executed in Kali Linux to identify running services, gather target information, and determine the target's operating system. The results depend on the target's responses and network conditions.